

Unit 6: From Farm to Table

NYS content module: Food Systems and Production (FSP) **Length:** 24 class periods of 40 minutes (days 144 to 167 of the year, inside marking period 4) **CTE themes carried:** Sustainability (primary); Financial and Consumer Literacy; Health, Safety, and Wellness; Career and Community Opportunities; Problem Solving and Innovation; Communication and Interpersonal Relationships **Unit project:** Feed the Class, scored with Rubric 06 - Feed the Class.md **Companion files in this folder:** Calendar 06.md, Vocabulary 06.md, Lessons 6.1 to 6.18, Project 06 - Feed the Class.md, Rubric 06 - Feed the Class.md, Assessment 06.1 - Food Systems Quiz.md (given inside Lesson 6.5 Day 2), Assessment 06.2 - Cooking Methods Check.md (given inside Lesson 6.11), Assessment 06.3 - Unit 6 Test.md (given inside Lesson 6.18 Day 2). Labs use Rubric 01 - Lab Rubric.md from Unit 1. Handouts and slide outlines are in 05 Materials/Unit 06/.

Reconciliation note: this plan was written to the scope and sequence and then reconciled to every lesson, the project, the rubric, and all three assessments as they landed in the folder. Where a lesson page and this plan disagree, the lesson page wins; the calendar and this plan follow the lessons' day counts, handout names, and check slots.

1. Why this unit

In my classroom I ran a multi-kitchen cooking project priced from a real supermarket circular. Four kitchens in one period, 112 students across four periods, \$150, eleven items, every price checked against the store's own weekly ad. I wrote six costed menus, a one-page master flowchart in five-minute blocks, and a station card for each kitchen. It worked, and the reason it worked was that all of the planning was on paper before anybody turned on a burner. But I did all of that planning. The students cooked.

Before I taught, I managed food and beverage at a hotel. That is where I learned what a food cost actually is: the number you have to hit every week, the difference between what you planned to spend and what the register said, and the fact that somebody has to explain that difference out loud. Nobody teaches a twelve-year-old that a meal has a budget and a schedule and a supply chain behind it. They teach a recipe.

So this unit hands the planning to the students. First they trace where food comes from, count what our own room throws away, and look at who in this county cannot reach a supermarket. Then they cook four times, once as a science experiment with a variable and a control. Then they run my system themselves: pick a dish, scale it from four servings to twenty-eight, price it from this week's circular, build their own flowchart, write their own station card, hand me one shopping list that I actually buy from, cook it, serve everybody else before they serve themselves, clean the room to a signed checklist, and tell me what it really cost per person and what they would change.

I want them to walk out knowing that a meal is a supply chain, a budget, and a plan, not just a recipe. And I want them to know that the jobs in that chain are real, that several of them are twenty-five miles from here, and that two of them you can start at sixteen.

2. Content identified (the concept map, written out)

2.1 What the NYS module lists for this unit

All 25 lines of the Food Systems and Production module are taught in this unit. None are skipped. Every line is printed on at least one lesson page.

A note on FSP 1. Basic Culinary Skills. Lines FSP 1. b) to e) duplicate N&W 1. a) to d) almost word for word (tools and equipment, safety and sanitation, accurate measuring, prepare a simple recipe). Unit 1 taught all four first, in Lessons 1.5, 1.1 to 1.4, 1.6, and 1.8. **This unit revisits them rather than introduces them.** That is on purpose and it is what the crosswalk recommends: the standards map lists these four lines under Unit 6 as the FSP module's own lines, with Unit 1 named as the first teaching. What changes in Unit 6 is the scale and the independence. In Unit 1 a student measured a quarter cup into a mug; in Unit 6 a student scales a recipe by seven, rounds to purchasable packages, and measures for thirty portions off a card the team wrote itself.

Line	Text (verbatim)	Taught in
FSP 1. Basic Culinary Skills a)	Apply basic culinary and management skills to the preparation of nutritious food	Lessons 6.6, 6.8, 6.14, 6.16, 6.17
FSP 1. b)	Identify common kitchen tools, equipment, and the uses for each	First taught Unit 1 Lesson 1.5. Revisited in Lesson 6.6 (tool for each method)
FSP 1. c)	Demonstrate safety and sanitation procedures when handling food and tools	First taught Unit 1 Lessons 1.1 to 1.4. Revisited in Lessons 6.8, 6.12 (the constraint card), 6.14 (the card's safety reminders), 6.16 (pre-flight and the station check)
FSP 1. d)	Demonstrate accurate measuring of dry/solid, liquid, and small quantity ingredients	First taught Unit 1 Lesson 1.6. Revisited and scaled up in Lessons 6.7 (the four batches) and 6.13 (scale by 7, convert units, round to packages)
FSP 1. e)	Prepare a simple recipe	First taught Unit 1 Lesson 1.8. Revisited in Lessons 6.7, 6.8, 6.9, and 6.16 (the same skill for 28 people off a student-written card)
FSP 2. Food Preparation a)	Identify specific foods from the region being studied	Lessons 6.2, 6.10

FSP 2. b)	Explore factors affecting regional food choices	Lessons 6.2, 6.10
FSP 2. c)	Prepare foods from the region being studied using ingredients common to the region	Lesson 6.9
FSP 2. d)	Demonstrate the skills and techniques common to food preparation of the region	Lessons 6.6, 6.7, 6.11
FSP 2. e)	Practice safety and sanitation practices and procedures in the preparation of food	Lessons 6.9, 6.16
FSP 3. Current Issues Related to Food in Global Societies a)	Describe factors affecting the food supply, including but not limited to geography, climate, economics, transportation systems, farming methods, energy, population density, political systems, natural disasters, and food waste	Lessons 6.1, 6.3
FSP 3. b)	Identify how current laws, regulations, and policies affect the availability, selection, and preparation of food in various regions and countries	Lesson 6.4 (SNAP, WIC, and food policy; printed on the lesson page and cited on objective 3) and Lesson 6.5 (packaging and waste rules)
FSP 3. c)	Explore factors that create current food-related health concerns around the world	Lesson 6.4
FSP 4. Future of Food in a Global Society a)	Understand the components of sustainable food systems	Lessons 6.1, 6.3, 6.5
FSP 4. b)	Explore the advantages and disadvantages of selecting and purchasing sustainable food products	Lesson 6.2
FSP 4. c)	Evaluate the quality of foods produced through sustainable methods	Lesson 6.5
FSP 4. d)	Identify the consequences of food scarcity and hunger	Lesson 6.4

	on society	
FSP 4. e)	Explore technological advances that affect the global food supply	Lesson 6.5
FSP 5. Consumer Resources and Finance a)	Demonstrate management of individual and family resources to select and purchase food	Lessons 6.12, 6.13, 6.15
FSP 5. b)	Analyze decisions about providing safe and nutritious food for individuals and families	Lesson 6.12 (printed on the lesson page and cited on objective 3), Project 06 (standards block), and Assessment 06.3 item 20
FSP 5. c)	Apply consumer skills to selecting and purchasing of food items	Lessons 6.4, 6.12, 6.13, 6.15
FSP 5. d)	Apply time management, organizational, and process skills to prioritize tasks and achieve goals	Lessons 6.8, 6.14, 6.17
FSP 6. Career Pathways a)	Explain roles and functions of individuals engaged in food industry careers	Lesson 6.18
FSP 6. b)	Investigate education, training requirements, and opportunities for career paths in global food industry	Lesson 6.18
FSP 6. c)	Assess personal employability skills for careers in the global foods industry and evaluate personal suitability for said careers	Lesson 6.18

Four N&W lines are also performed in this unit because the Unit 6 content is where they get their real use: **N&W 1. h) Demonstrate appropriate table setting** (Lesson 6.11, performed for 30 covers in Lesson 6.17), **N&W 1. i) Practice food sharing behaviors that show consideration for other cultures and promote social acceptance** (Lessons 6.10, 6.11, 6.17), **N&W 1. g) Prepare and serve a simple meal** (Lesson 6.17), and **N&W 3. a) Create and organize a shopping list** (Lesson 6.15, at class scale). All four were first taught in Unit 1.

2.2 How the CTE themes come in

- **Sustainability (primary):** the whole first topic is built on it. SUS 1. Resources and SUS 2. Environmental Management carry Lessons 6.1 to 6.5: the supply chain trace, food miles, our own waste audit with a real number on the wall, composting, crop rotation, and a packaging redesign. The theme comes back on the last day of the project when Zone 4 counts what the class threw away and posts the number next to the Lesson 6.3 audit number.
- **Financial and Consumer Literacy:** FCL 4. Buying Goods and Services runs the whole project. Students develop a spending plan against a \$20 station budget and an \$80 class budget (4. c)), compare local and national sources and store brands (4. g)), apply a decision model to a purchase (4. f)), and demonstrate consumer skills in a real purchase the teacher actually makes (4. i)).
- **Health, Safety, and Wellness:** HSW 4. Classroom and Workplace Safety is the thread that never gets cut. Four labs and two cook days run on the same six-step sanitation routine, the same constraint card, the same station check, and the same rule that only the teacher opens an oven or drains a pot.
- **Career and Community Opportunities:** CCO 4. Career Plans and CCO 7. Career Pathways carry Lesson 6.18, thirteen jobs across eight links of the chain, five training paths, and ServSafe as a credential a high school student can actually earn. CCO 1. The Work World b) also appears in Lesson 6.1, because the supply chain trace is a list of jobs.
- **Problem Solving and Innovation:** PSI 1. Problem Solving a) is the quick bread experiment, one variable and four constants. PSI 2. Design Process a) c. is the constraint card in Lesson 6.12 and a) f. is the packaging prototype in Lesson 6.5. PSI 1. e) is the whole project: a real problem with real limits and one shot at it.
- **Communication and Interpersonal Relationships:** CIR 3. Workplace Communication f) is every lab team and every station on cook day; CIR 3. e) is the station card, which is a written report another team has to be able to act on. CIR 6. f) is the table in Lessons 6.10, 6.11, and 6.17, where the rule is that everyone's food and everyone's tradition gets respect.

2.3 Trends, issues, and news

1. **Grocery prices.** Students hear adults talk about what food costs. This unit hands them the circular and makes them do the arithmetic. The number they end up with, about \$2.70 to \$2.80 a person for a full meal, is the most useful number in the unit, because it is theirs.
2. **Delivery apps.** Most students have watched an adult order food to the door. Lesson 6.1's chain gets one more link in the debrief, and Lesson 6.18's logistics coordinator card is the job that makes that link work. The question to ask: who pays for the driver, and is it in the price on the screen?
3. **Plant-based products.** Meat alternatives are in every supermarket now. Lesson 6.2 puts them in the local versus global discussion and Lesson 6.5 puts them in the technology discussion (FSP 4. e)). Two of the five project menu options are meatless, and that is not framed as a sacrifice.

4. **Local food and school gardens.** Farmers markets, CSA boxes, and school gardens are visible on Long Island. Lesson 6.5 plants something that is still alive on the day of Lesson 6.11 and is watered to the end of the unit, which is weeks of care students can see.
5. **Food waste rules for businesses.** New York State requires large food scrap generators to donate excess edible food and recycle food scraps where a processor is within range. That is a real law that connects Lesson 6.3's audit to Lesson 6.4's pantries and to FSP 3. b).
6. **Climate and crops.** Students have seen a drought or a flood story affect a price. Lesson 6.1's list of factors affecting the food supply is the state's own list and it includes climate, natural disasters, and transportation. Use whatever is in the news that month.

2.4 What students this age find interesting

1. Cooking something they get to eat, in a room that smells like food.
2. Real money, spent on them, that they got to decide about.
3. Being in charge of a station, with a title and a job nobody else has.
4. An experiment where one thing changes and they get to find out what happens.
5. Food from their own family's kitchen being named out loud in school and treated as worth knowing.

2.5 Links to other subjects

- **Math:** ratio and the scale factor (6.13), unit conversion between cups, fluid ounces, and ounces by weight (6.13), unit price and unit rate (6.13, 6.15), averages and percent (6.3's waste data, 6.17's variance as a percent of plan). The four costing sheets are a ready problem set.
- **Science:** heat transfer and why dry heat browns and moist heat does not (6.6), the acid and base reaction that makes baking soda work and the complete reaction in baking powder (6.7), food spoilage and the danger zone (6.3, 6.16), and plant growth in the windowsill planting (6.5).
- **Social studies:** trade, migration, and cuisine. Lesson 6.10 takes one dish and finds it in three cultures, which is the story of trade routes and of people moving. The spice trade and the movement of staple crops between continents are the same content the social studies course teaches as history.
- **ELA:** procedural writing. The station card in Lesson 6.14 is a technical document with a test attached: another team has to be able to follow it. The dish research card in Lesson 6.10 is a short research summary with a source named. The reflection in Lesson 6.17 is a claim with evidence from the student's own documents.
- **Art:** plating and the display. Lesson 6.11's place setting and Lesson 6.17's 30 covers use composition, spacing, and the one-inch line. The menu board and the packaging redesign in Lesson 6.5 use layout and color to make something read from across the room.

- **Health:** the nutrition content stays in Unit 1 and in the health course. This unit's overlap is food safety, food access, and the cultural and social side of eating. See Section 11.

3. Enduring understandings and essential questions

Big ideas:

- Every bite of food traveled through a chain of people and places, and every link in that chain is somebody's job.
- Feeding a group is a budget and a schedule before it is a recipe. In a kitchen, time is the ingredient you cannot buy more of.
- One step always holds up everything else. Find it and build the clock around it.
- The choices a household or a school makes about food have costs beyond the price tag: fuel, water, soil, packaging, and what gets thrown away.
- Not everyone can reach the same food for the same price, and that is a system, not an accident.
- Cooking is chemistry you can eat. Change one thing on purpose and you can find out what it does.

Essential questions:

1. Where did this food come from, and who touched it before I did?
2. Can we feed this whole class on what I can spend, in the time we have, with the rules this room has?
3. What is the one step everything waits on?
4. What does our food actually cost, and what does it cost that is not on the receipt?
5. Who in this county cannot reach the food we just priced, and why?
6. Which job in the food chain would I actually want, and what is the shortest honest path to it?

4. Knowledge and skills

Knowledge (facts, vocabulary, ideas):

- The links in a food supply chain: producer, processor, distributor, retailer, consumer, and the cold chain that runs through all of them.
- The state's own list of factors affecting the food supply: geography, climate, economics, transportation systems, farming methods, energy, population density, political systems, natural disasters, and food waste.
- Local, seasonal, in season and out of season, food miles, farmers market, CSA, storage crop.
- Food insecurity, food desert, food pantry, SNAP, WIC, community garden, mobile market.
- Sustainability, compost, crop rotation, drip irrigation, integrated pest management, packaging, single use, shelf life; the difference between food waste and food loss.

- Dry heat and moist heat methods and what each one does to food: bake, roast, broil, toast, sauté; boil, simmer, steam, poach. Browning happens with dry heat.
- Leavening: baking powder, baking soda, the acid it needs, and the carbon dioxide that makes a quick bread rise. Variable, control, hypothesis.
- Cuisine, staple crop, flatbread, filled dough, legume, tradition, and why the same dish shows up in three cultures.
- Place setting, cover, flatware, host, shared platter, hospitality.
- Yield, portion, per serving, scale factor, purchasable unit, unit price, store brand, food cost.
- Flowchart, critical path, station card, mise en place, hold time, station check.
- The seven project constraints and the six-step sanitation routine.
- The thirteen food-chain careers, the eight links, the five training paths, and what ServSafe is.

Skills (what students will be able to do):

- Trace one real food from farm to plate and name the job at each link.
- Read a New York seasonal chart and build two meals, one local and seasonal and one not, and compare them on price and food miles.
- Collect and record waste data from their own room, compute a total, and scale it to a week and a year.
- Read a neighborhood map and a price comparison and decide whether an area is a food desert.
- Build a packaging prototype that uses less material, and defend it.
- Sort nine cooking methods into dry heat and moist heat and name the tool and the right food for each.
- Run a four-batch experiment with one variable and four constants, predict, measure the rise, taste, and write a conclusion.
- Prepare and serve a pasta dish and a quesadilla dish at a station of seven inside a 40-minute period, on role cards, against the lab rubric.
- Research one dish in three cultures on a card with a source, and present it.
- Set a correct place setting and say why each piece is where it is.
- Compute a scale factor, multiply every ingredient, convert between cups, fluid ounces, and ounces by weight, round up to purchasable packages, and run a yield check against 30 portions.
- Cost a scaled recipe from a real circular with a running total, compute unit prices, find and log a store-brand swap with the dollar amount saved, and revise a plan that came in over budget.
- Build an eight-block, five-minute flowchart with a role named on every line, identify the critical path, and predict the effect of a five-minute slip as a real time.
- Write a station card with ten one-action steps, an allergen box, safety reminders, and four visible “done looks like” checks that another team can follow with no questions.
- Consolidate four costing sheets into one class shopping list sorted by store section, with a swap on every line, and pass a twelve-item approval checklist.

- Run a pre-flight check, execute a role in every block of a cook day, call the teacher at the right minute, serve 30 covers others-first, and pass a signed station check.
- Compute actual versus planned cost per person from a real receipt and explain the difference.
- Evaluate their own project against their own plan and name one change tied to a specific block, line, or step.
- Place thirteen careers on the chain and sort them by education needed, and assess personal fit with a reason.

5. Unit objectives

One per topic, upper half of Bloom's:

- **Topic 6.1:** Students will analyze where their food comes from by tracing a food from farm to plate, auditing their own room's waste, and evaluating a neighborhood's food access, then justify one sustainable practice the school could adopt.
- **Topic 6.2:** Students will apply dry heat and moist heat methods in four lab settings, design and run a controlled leavening experiment with one variable, and evaluate how culture shapes the same dish in three places.
- **Topic 6.3:** Students will design and execute a complete meal for the whole class by scaling a recipe, costing it from a real circular against a \$20 station budget, building a five-minute-block flowchart with a named critical path, writing a station card another team could follow, cooking and serving it inside the period, and evaluating the plan against what actually happened and what it actually cost.
- **Topic 6.4:** Students will evaluate thirteen careers across the food chain against their own interests and the training each requires, and justify which one they would and would not want.

6. Topics, sequence, and timeline

Topic	Days	Lessons	Activity lines (verb first)	Formal check at end of topic
6.1 Where Food Comes From	6 (days 144 to 149)	6.1 Trace a Food From Farm to Plate (1); 6.2 Local, Seasonal, and Global (1); 6.3 Food Waste Audit in Our Room (1); 6.4 Food Scarcity and Food Deserts (1); 6.5 Sustainable Food Methods	trace one real food across ten steps and name the job at each; sort foods by season on the New York chart; build two meals, one local and seasonal and one not, and compare price and food miles; collect	Assessment 06.1 - Food Systems Quiz.md, 12 items, 21 points, in the last 15 minutes of Lesson 6.5 Day 2

		and the Food Systems Quiz (2)	and weigh or tally the room's food waste and scale it to a week and a year; map a neighborhood's food access and compare prices at a corner store and a supermarket; sort compost from trash; plant seeds in cups for the windowsill; redesign a package to use less material and defend it	
6.2 Food Preparation Methods and Science	8 (days 150 to 157)	6.6 Cooking Methods: Dry Heat, Moist Heat, and Why (1); 6.7 Quick Bread Science Lab (2); 6.8 Lab: Cheesy Pasta Marinara With Role Cards (1); 6.9 Lab: Black Bean Quesadillas (1); 6.10 Global Foods: One Dish, Three Cultures (2); 6.11 Table Setting and Food Sharing Across Cultures and the Methods Check (1)	sort nine methods into dry heat and moist heat and name the tool for each; watch the teacher demonstration and record what changed; write a hypothesis for four leavening treatments; mix, bake, measure the rise, taste, and conclude; prepare and serve cheesy pasta marinara on role cards inside 40 minutes;	Assessment 06.2 - Cooking Methods Check.md, 12 items, 21 points, in the last 15 minutes of Lesson 6.11. Labs 6.7, 6.8, and 6.9 are each scored on Rubric 01 - Lab Rubric.md (16 points)

			prepare and serve black bean quesadillas and measure the assembly line's throughput; research one dish in three cultures on a card with a source and present it; set a correct place setting and explain each piece; write the class food sharing agreement	
6.3 Feed the Class project	8 (days 158 to 165)	6.12 Feed the Class: Menu Options and Constraints (1); 6.13 Scale the Recipe and Cost It From the Circular (2); 6.14 Build the Master Flowchart (1); 6.15 Shopping List and Unit Price Check (1); 6.16 Cook Day (2); 6.17 Serve, Eat, Clean, and Reflect (1)	list the seven constraints and what each rules out; compare five costed menu options on cost per person, difficulty, and equipment; select an option and justify it against three constraints; compute the scale factor and multiply every ingredient; convert cups to fluid ounces to ounces by weight; round up to purchasable packages; run	Rubric 06 - Feed the Class.md, five criteria, 20 points, from the folder collected at the end of Lesson 6.17. Cook day is also scored individually on Rubric 01 - Lab Rubric.md

			the yield check against 30; cost every line from the circular with a running total; log a store-brand swap with the dollar amount saved; revise a plan that came in over budget; build an eight-block flowchart with a role on every line; identify the critical path and predict a five-minute slip as a real time; write a ten-step station card with an allergen box; consolidate four sheets into one list sorted by store section; pass a twelve-item approval checklist; run the pre-flight and the six-step sanitation routine; cook, portion, and serve 30 covers inside the period; set 30 covers and serve others first; clean by four zones and	
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			count the waste; compute actual versus planned cost per person and explain the difference; name one change tied to a block, a line, or a step	
6.4 Careers	2 (days 166 to 167)	6.18 Careers in Food Systems, and the Unit 6 Review and Test (2)	place thirteen careers on the eight-link chain; re-sort them by education needed into four columns; name the earliest start age for each of the five training paths; complete the “would I like this” self-check with a reason; circle the jobs that exist within 25 miles; play Four Links as a team review; take the unit test	Assessment 06.3 - Unit 6 Test.md, 20 items, 40 points, in the last 20 minutes of Lesson 6.18 Day 2. Topic 6.4 has no separate check; it is assessed inside the unit test

Day counts: $6 + 8 + 8 + 2 = 24$. The day-by-day placement is in Calendar 06.md.

Flex lessons (drop first when time runs short), in this order:

1. **Lesson 6.10 Day 2** (fold the presentations into Day 1 as a two-minute gallery share instead of full presentations; the research card is still written and still scored). Saves 1 day.
2. **Lesson 6.2** (fold local and seasonal into Lesson 6.1’s debrief and send the New York seasonal chart home as the unit’s one homework sheet; FSP 2. a), 2. b), and 4. b) still get taught in 6.10 and 6.12). Saves 1 day.

3. **Lesson 6.5 Day 2** (do the packaging redesign as a sketch instead of a prototype, and move the Food Systems Quiz to the last 15 minutes of Lesson 6.4). Saves 1 day.
4. **Lesson 6.7 Day 1** (run the acid-and-base fizz test as a five-minute teacher demonstration at the start of Day 2 and give the hypothesis sheet as the do now). Saves 1 day, and costs the most in learning of the four.

Lessons that must not be cut: 6.14 (the flowchart; without it cook day fails and there is nothing to reflect against), 6.15 (the approval checklist; without it there is no food), 6.16 (cook day itself), and 6.4 (food deserts, which is the lesson that answers why any of this matters to somebody who is not going to be a chef). Also do not cut the safety content inside 6.8, 6.9, and 6.16; a lab without the six-step routine and the station check is not a shorter lab, it is a different one.

The 21-day version (for the 30-week schedule, which the scope and sequence sets at 21 days for this unit): drop flex lessons 1, 2, and 3 above. Topic 6.1 becomes 4 days (6.1 with local and seasonal folded in, 6.3, 6.4 with the quiz in the last 15 minutes, 6.5 in one day), Topic 6.2 becomes 7 days (6.10 in one day), Topic 6.3 stays at 8, Topic 6.4 stays at 2. Total 21. The project is untouched, which is the point: it is the thing a principal will ask about.

The 10-day version (for the one-semester or every-other-day schedule, which the scope and sequence sets at 10 days):

Day	Lesson	What survives
1	6.1 with 6.2 folded in	The supply chain trace, plus local and seasonal as the debrief
2	6.3	The waste audit, with the tally version instead of the scale
3	6.4	Food deserts and the neighborhood map
4	6.6	Dry heat and moist heat with the teacher demonstration
5	6.9	One lab, the quesadillas, because it is the cheapest and the most forgiving
6	6.11	Place setting and the food sharing agreement, with the Methods Check as a 10-item version
7	6.12 and 6.13 Day 1 combined	Constraints, the menu choice, the scale factor, the yield check

8	6.13 Day 2 and 6.15 combined	Costing from the circular and the one class shopping list with the approval checklist
9	6.14 and 6.16 combined	The flowchart written at the start of the period, then one cook day with all stations cooking at once
10	6.17	Serve, clean, the cost comparison, and the four-question reflection. The unit test is the do now of the next unit's first day, or the reflection replaces it

In the 10-day version the careers content moves to Unit 7, where the thirteen cards become part of the career research. Say so in the syllabus, because FSP 6. a), b), and c) still need a home.

Every-other-day version (a year of 90 meetings): the 10-day plan above, except that the growing in Lesson 6.5 is dropped (nothing to water on the off days) and the cook day gets two meetings instead of one, for 11 meetings total.

7. Standards block

NYS module lines taught (every line, by code, with the lesson that teaches it)

Text verbatim from NYS Middle Level CTE FACS Content Module 06, Food Systems and Production (nyc-tecenter.org, June 2018 preliminary release).

- NYS FSP 1. Basic Culinary Skills a) Apply basic culinary and management skills to the preparation of nutritious food : Lessons 6.6, 6.8, 6.14, 6.16, 6.17
- NYS FSP 1. Basic Culinary Skills b) Identify common kitchen tools, equipment, and the uses for each : Unit 1 Lesson 1.5 (first taught); Lesson 6.6 (revisited)
- NYS FSP 1. Basic Culinary Skills c) Demonstrate safety and sanitation procedures when handling food and tools : Unit 1 Lessons 1.1 to 1.4 (first taught); Lessons 6.8, 6.12, 6.14, 6.16 (revisited)
- NYS FSP 1. Basic Culinary Skills d) Demonstrate accurate measuring of dry/solid, liquid, and small quantity ingredients : Unit 1 Lesson 1.6 (first taught); Lessons 6.7, 6.13 (revisited and scaled)
- NYS FSP 1. Basic Culinary Skills e) Prepare a simple recipe : Unit 1 Lesson 1.8 (first taught); Lessons 6.7, 6.8, 6.9, 6.16 (revisited)
- NYS FSP 2. Food Preparation a) Identify specific foods from the region being studied : Lessons 6.2, 6.10
- NYS FSP 2. Food Preparation b) Explore factors affecting regional food choices : Lessons 6.2, 6.10

- NYS FSP 2. Food Preparation c) Prepare foods from the region being studied using ingredients common to the region : Lesson 6.9
- NYS FSP 2. Food Preparation d) Demonstrate the skills and techniques common to food preparation of the region : Lessons 6.6, 6.7, 6.11
- NYS FSP 2. Food Preparation e) Practice safety and sanitation practices and procedures in the preparation of food : Lessons 6.9, 6.16
- NYS FSP 3. Current Issues Related to Food in Global Societies a) Describe factors affecting the food supply, including but not limited to geography, climate, economics, transportation systems, farming methods, energy, population density, political systems, natural disasters, and food waste : Lessons 6.1, 6.3
- NYS FSP 3. Current Issues Related to Food in Global Societies b) Identify how current laws, regulations, and policies affect the availability, selection, and preparation of food in various regions and countries : Lessons 6.4, 6.5
- NYS FSP 3. Current Issues Related to Food in Global Societies c) Explore factors that create current food-related health concerns around the world : Lesson 6.4
- NYS FSP 4. Future of Food in a Global Society a) Understand the components of sustainable food systems : Lessons 6.1, 6.3, 6.5
- NYS FSP 4. Future of Food in a Global Society b) Explore the advantages and disadvantages of selecting and purchasing sustainable food products : Lesson 6.2
- NYS FSP 4. Future of Food in a Global Society c) Evaluate the quality of foods produced through sustainable methods : Lesson 6.5
- NYS FSP 4. Future of Food in a Global Society d) Identify the consequences of food scarcity and hunger on society : Lesson 6.4
- NYS FSP 4. Future of Food in a Global Society e) Explore technological advances that affect the global food supply : Lesson 6.5
- NYS FSP 5. Consumer Resources and Finance a) Demonstrate management of individual and family resources to select and purchase food : Lessons 6.12, 6.13, 6.15
- NYS FSP 5. Consumer Resources and Finance b) Analyze decisions about providing safe and nutritious food for individuals and families : Lesson 6.12, Project 06, Assessment 06.3 item 20
- NYS FSP 5. Consumer Resources and Finance c) Apply consumer skills to selecting and purchasing of food items : Lessons 6.4, 6.12, 6.13, 6.15
- NYS FSP 5. Consumer Resources and Finance d) Apply time management, organizational, and process skills to prioritize tasks and achieve goals : Lessons 6.8, 6.14, 6.17
- NYS FSP 6. Career Pathways a) Explain roles and functions of individuals engaged in food industry careers : Lesson 6.18
- NYS FSP 6. Career Pathways b) Investigate education, training requirements, and opportunities for career paths in global food industry : Lesson 6.18
- NYS FSP 6. Career Pathways c) Assess personal employability skills for careers in the global foods industry and evaluate personal suitability for said careers : Lesson 6.18

N&W lines performed in this unit (all first taught in Unit 1):

- NYS N&W 1. Culinary Management g) Prepare and serve a simple meal : Lesson 6.17
- NYS N&W 1. Culinary Management h) Demonstrate appropriate table setting : Lessons 6.11, 6.17
- NYS N&W 1. Culinary Management i) Practice food sharing behaviors that show consideration for other cultures and promote social acceptance : Lessons 6.10, 6.11, 6.17
- NYS N&W 3. Consumer Resources and Finance a) Create and organize a shopping list : Lesson 6.15

CTE theme lines used (text verbatim from the theme modules)

Sustainability (primary):

- CTE SUS 1. Resources a) Define “sustainability” as it applies to resource use : Lesson 6.5
- CTE SUS 1. Resources e) Investigate practices that promote stewardship of environmental resources : Lessons 6.3, 6.17
- CTE SUS 2. Environmental Management c) Describe ways individuals can adjust behaviors to minimize the impact of their choices on the environment : Lesson 6.3
- CTE SUS 2. Environmental Management h) Explain factors to consider when assessing the environmental impact of food choices : Lessons 6.1, 6.2
- CTE SUS 4. Careers Related to Sustainability a) Investigate knowledge, skills, and practices needed for careers related to sustainability or for employment in settings that utilize sustainable practices : Lesson 6.18

Financial and Consumer Literacy:

- CTE FCL 4. Buying Goods and Services c) Develop a spending and savings plan (budget) based on income and expenses : Lesson 6.12
- CTE FCL 4. Buying Goods and Services f) Apply a decision-making model to maximize consumer satisfaction when purchasing goods and services : Lesson 6.13
- CTE FCL 4. Buying Goods and Services g) Compare local, national, and global resources for obtaining goods and services : Lessons 6.2, 6.4, 6.13, 6.15
- CTE FCL 4. Buying Goods and Services i) Demonstrate personal development of consumer skills through practice of these skills in a variety of classroom applications : Lesson 6.15

Health, Safety, and Wellness:

- CTE HSW 4. Classroom and Workplace Safety c) Apply safety practices when using tools, machines, and equipment : Lessons 6.6, 6.8, 6.9, 6.16
- CTE HSW 4. Classroom and Workplace Safety e) Participate in housekeeping procedures that help to establish and maintain safe working conditions in the classroom or workplace : Lessons 6.8, 6.9, 6.16, 6.17

Career and Community Opportunities:



- CTE CCO 1. The Work World b) Discuss the value of work as it relates to the individual and to society : Lesson 6.1
- CTE CCO 4. Career Plans a) Examine potential career choices to determine knowledge, skills, and abilities associated with each : Lesson 6.18
- CTE CCO 5. Community Needs Assessment b) Research community resources available to support individuals and families : Lesson 6.4
- CTE CCO 7. Career Pathways c) Research ways social, economic, and technological changes have led to adaptations in work and community environments and expectations : Lesson 6.18

Problem Solving and Innovation:

- CTE PSI 1. Problem Solving a) Describe the scientific method of inquiry as it relates to real-world problem solving : Lessons 6.6, 6.7
- CTE PSI 1. Problem Solving e) Demonstrate personal development of problem-solving skills through practice of these skills in a variety of classroom applications : Lesson 6.7
- CTE PSI 2. Design Process (Proactive) a) c. Defining constraints that must be adhered to : Lesson 6.12
- CTE PSI 2. Design Process (Proactive) a) f. Developing and constructing a prototype or model of the selected design : Lesson 6.5

Communication and Interpersonal Relationships:

- CTE CIR 3. Workplace Communication e) Explain how to give and receive accurate reports in a variety of formats : Lesson 6.14
- CTE CIR 3. Workplace Communication f) Demonstrate effective communication skills in a group setting to accomplish a task : Lessons 6.8, 6.9, 6.14, 6.16
- CTE CIR 6. Conflict Prevention and Management d) Explore how similarities and differences among people affect conflict prevention and management : Lessons 6.10, 6.11
- CTE CIR 6. Conflict Prevention and Management f) Contribute to a classroom environment that encourages respect for the ideas, perspectives, and contributions of all : Lessons 6.10, 6.11, 6.17

NYS FACS Learning Standards (Intermediate)

The state's FSP module lists Standards 1, 2, and 3. This unit cites Standards 1 and 3 on lesson pages. Standard 2 is carried at unit level: the key idea's sentence about maintaining a safe workplace is what the constraint card, the pre-flight, and the station check are, and no Standard 2 performance indicator fits a food lab better than Standard 1 PI B does, so the lessons cite Standard 1 PI B and this plan records Standard 2 at the key idea level.

Standard 1: Personal Health and Fitness. Students will have the necessary knowledge and skills to establish and maintain physical fitness, participate in physical activity, and maintain personal health. Key Idea: Students will use an understanding of the elements of good nutrition to plan appropriate diets for themselves and others. They will know and use the appropriate tools and technologies for safe and healthy food preparation.

- PI A: Students understand the relationships among diet, health, and physical activities; evaluate their own eating patterns; and use appropriate technology and resources to make food selections and prepare simple, nutritious meals. (Lessons 6.2, 6.6, 6.7, 6.8, 6.9, 6.10, 6.11, 6.17)
- PI B: Students apply principles of food safety and sanitation. (Lessons 6.7, 6.8, 6.9, 6.16)
- PI C: Students recognize the mental, social, and emotional aspects of good health. (Lessons 6.10, 6.11)

Standard 3: Resource Management. Students will understand and be able to manage their personal and community resources. Key Idea: Students will understand and be able to manage personal resources of talent, time, energy, and money and make effective decisions in order to balance their obligations to work, family, and self.

- PI A: Students understand how the family can provide for the economic, physical, and emotional needs of its members. (Lesson 6.4)
- PI B: Students understand the resources available to them, make informed decisions about the use of those resources, and know some ways to expand resources. (Lessons 6.1, 6.3, 6.5, 6.15)
- PI C: Students are able to budget their time and money. (Lessons 6.12, 6.13, 6.14)
- PI E: Students identify their own abilities and interests as possible guides to career choice. (Lesson 6.18)

Standard 2: A Safe and Healthy Environment. Students will acquire the knowledge and ability necessary to create and maintain a safe and healthy environment. Key Idea: Students will know the basic principles of home and community safety. They can demonstrate the skills necessary to maintain their homes and workplaces in a safe and comfortable condition. They can provide a safe and nurturing environment for themselves and others. (Unit-level citation for the constraint card, the pre-flight, and the station check in Lessons 6.12, 6.14, 6.16, and 6.17; the fit is the key idea's second sentence, and no Standard 2 PI applies, as the crosswalk notes.)

The PI letters are NYLearns codes; the NYSED print document lists the indicators as bullets.

CDOS (Intermediate)

- **CDOS 1 Career Development:** Students will be knowledgeable about the world of work, explore career options, and relate personal skills, aptitudes, and abilities to future career decisions. (Lesson 6.18)
- **CDOS 2 Integrated Learning:** Students will demonstrate how academic knowledge and skills are applied in the workplace and other settings. (Lessons 6.13, 6.16; the scaling, the unit conversion, and the unit price are math doing real work)
- **CDOS 3a Universal Foundation Skills:** Students will demonstrate mastery of the foundation skills and competencies essential for success in the workplace. Sub skills used in this unit:
 - **Basic Skills:** Students listen to and read the ideas of others and analyze what they hear and read; acquire and use information from a variety of sources; and

apply a combination of mathematical operations to solve problems in oral or written form. (Lessons 6.3, 6.13, 6.15)

- **Thinking Skills:** Students evaluate facts, solve advanced problems, and make decisions by applying logic and reasoning skills. (Lessons 6.2, 6.4, 6.7)
- **Personal Qualities:** Students demonstrate an understanding of the relationship between individuals and society and interact with others in a positive manner. (Lessons 6.11, 6.16)
- **Interpersonal Skills:** Students demonstrate the ability to work with others, present facts that support arguments, listen to dissenting points of view, and reach a shared decision. (Lesson 6.17; the lab teams at unit level)
- **Technology:** Students select and use appropriate technology to complete a task. (Lesson 6.6; tools and equipment)
- **Managing Information:** Students select and communicate information in an appropriate format (e.g., oral, written, graphic, pictorial, multimedia). (Lessons 6.10, 6.14)
- **Managing Resources:** Students understand the material, human, and financial resources needed to accomplish tasks and activities. (Lessons 6.8, 6.9, 6.12, 6.14, 6.15)
- **Systems:** Students understand the process of evaluating and modifying systems within an organization. (Lessons 6.1, 6.5)

Standard 3b (Career Majors) exists only at Commencement. A grade 6 to 8 course cites 1, 2, and 3a only, which is what all twelve state modules do.

National FCS 3.0 (LEAD FCS Education, 2018 to 2028; text verbatim from the 3.0 compilation)

The FSP module's own Standards Addressed block prints 3.0 wording for every line it cites, which makes this the cleanest module in the course to cite. Numbers printed in the FSP block are marked (block); numbers confirmed in the 3.0 text but not printed in the FSP block are marked (ext).

Core competencies for this unit:

- 14.1.2 Look into how psychology, culture, and social pressure shape food choices and nutrition habits. (block) (Lesson 6.10)
- 14.1.3 Look into how government, the economy, and technology shape food choices and practices. (block) (Lesson 6.4)
- 14.1.4 Examine how events and conditions, local to global, change what people eat. (block) (Lessons 6.2, 6.10)
- 14.1.5 Examine laws and regulations that touch nutrition and wellness. (block) (Lesson 6.4; the competency the crosswalk pairs with FSP 3. b))
- 14.3.3 Show the ability to select, store, prepare, and serve food that is nourishing and looks good on the plate. (block) (Lessons 6.8, 6.9, 6.11, 6.16, 6.17)
- 14.3.4 Judge policies and practices that affect food security, sustainability, food integrity, and family nutrition. (block) (Lessons 6.1, 6.4)

- 14.4.1 Examine the conditions and habits that keep food safe to eat. (block) (Lesson 6.16)
- 14.4.2 Examine safety and sanitation practice. (block) (Lessons 6.8, 6.16)
- 14.4.3 Examine how changes in food production and distribution, sustainability, organic farming, and genetically modified foods included, shape the food supply. (block) (Lesson 6.1)

Secondary competencies cited on lesson pages, confirmed in the 3.0 text:

- 2.1.1 Use time management, organization, and step-by-step process skills to rank tasks and reach goals. (ext; the state line FSP 5. d) is this competency word for word) (Lesson 6.14)
- 2.1.3 Examine the decisions behind providing food that is safe and nourishing for individuals and families. (ext; the state line FSP 5. b) is this competency word for word) (Lessons 6.12, 6.13, 6.15)
- 2.2.3 Practice habits that conserve, reuse, and recycle so resources and the environment last. (ext) (Lesson 6.5)
- 2.5.1 Examine how resources get used when individuals, families, and communities choose among needs and wants. (ext) (Lesson 6.2)
- 3.4.3 Try out different ways to save energy and cut waste. (ext) (Lesson 6.3)
- 6.2.4 Treat people of every age, gender, culture, and background with respect and fairness. (ext) (Lessons 6.10, 6.11)
- 8.1.1 Explain what people in food production and food service careers do. (ext) (Lesson 6.18)
- 8.1.3 Sum up the education and training that food production and service career paths require and where they lead. (ext) (Lesson 6.18)
- 8.4.2 Use menu-planning principles to build and adjust menus. (ext) (Lesson 6.12)
- 8.4.3 Work out the food, equipment, and supplies a menu needs. (ext) (Lesson 6.14)
- 8.4.7 Use measurement, portion control, conversions, food costing, menu terms, and pricing when planning a menu. (ext) (Lesson 6.13)
- 8.5.2 Cook with professional skill across a range of methods, from roasting and grilling to braising, poaching, steaming, and baking, on professional equipment. (ext) (Lesson 6.6)
- 8.5.3 Demonstrate knowledge of portion control and proper scaling and measurement techniques (ext) (Lesson 6.13)
- 8.5.7 Prepare fruits, vegetables, starches, legumes, dairy, fats, and oils safely and with professional technique. (ext) (Lessons 6.8, 6.9)
- 8.5.10 Prepare breads, baked goods, and desserts safely and with professional technique. (ext) (Lesson 6.7)
- 8.5.12 Plate, garnish, and present food to a professional standard. (ext) (Lessons 6.11, 6.17)
- 8.5.13 Build sustainability into food work, from menu planning and buying through preparation, service, storage, recycling, and waste. (ext) (Lessons 6.3, 6.5, 6.17)

- 8.5.14 Cook in ways that raise nutritional value, cut calories and fat, and use herbs and spices for flavor. (ext) (Lesson 6.6)
- 8.6.1 Use sound purchasing, receiving, and storage practice in a food service operation. (ext) (Lesson 6.15)
- 8.2.7 Show safe food handling and preparation that keeps hazardous foods from cross contaminating others. (ext) (Lesson 6.16)
- 9.1.3 Sum up the education and training that food science, dietetics, and nutrition career paths require. (ext) (Lesson 6.18)
- 9.5.6 Run sensory evaluations of food products. (ext) (Lesson 6.7)
- 14.5.1 Look into how advances in science and technology change the nutrients, availability, and safety of food. (ext; printed in the N&W block with older wording) (Lesson 6.5)
- 1.2.1 Examine possible careers to find what each one requires and offers in knowledge, skills, attitudes, and openings. (ext) (Lesson 6.18)

Not cited: 14.1.1 is printed in the FSP block but no lesson in this unit teaches it; the components of individual and family wellness are Unit 1's. 14.1.5 (legislation and regulations related to nutrition and wellness) is now cited on Lesson 6.4, where it carries FSP 3. b).

Area 14 has no career competency in the 3.0 text. Lesson 6.18 therefore cites 8.1.1, 8.1.3, and 9.1.3 for the food careers, which is the same substitution Lesson 1.22 made. Credit LEAD FCS Education visibly on the site; it is a condition of the non-commercial terms of use.

Career Ready Practices (as the FSP module lists them; wording from the extract's assembled list)

The FSP module PDF prints these eight:

- CRP 2. Apply appropriate and academic and technical skills
- CRP 4. Communicate clearly and effectively and with reason
- CRP 5. Consider environmental, social, and economic impacts of decisions
- CRP 6. Demonstrate creativity and innovation
- CRP 7. Employ valid and reliable research strategies
- CRP 8. Utilize critical thinking to make sense of problems and persevere in solving them
- CRP 9. Model integrity, ethical leadership, and effective management
- CRP 12. Work productively in teams while using cultural global competence

CRP 2 is printed "Apply appropriate and academic and technical skills" in the FSP PDF, which is the state's own typo; the N&W PDF prints the corrected "Apply appropriate academic and technical skills." The line above is quoted as the FSP module prints it. Career Ready Practices are a secondary citation on this unit page only; the CCTC source text has not been fetched (crosswalk E.2).

8. Project

Name: Feed the Class. Full sheet in Project 06 - Feed the Class.md (handed out in Lesson 6.12 and reread in Lesson 6.15); rubric in Rubric 06 - Feed the Class.md.

The scenario: twenty-eight people in the room and about eighty dollars. In eight class periods each station picks a dish, scales it from four servings to twenty-eight, prices it to the penny from this week's real supermarket circular, writes its own five-minute-block flowchart and its own ten-step station card, hands the teacher one consolidated shopping list that the teacher actually buys from, cooks it, serves the class before serving itself, cleans the room to a signed checklist, and reports what it really cost per person and what it would change. This is my own multi-kitchen system with the planning handed to the students.

Deliverables: (1) the scaled recipe with the math shown, including the unit conversions and the yield check against 30 portions; (2) the costed shopping list, the swap log, and the station's initialed rows on the approved class list; (3) the station flowchart, eight blocks, a role on every line, the critical path circled, the actual times written in after cook day; (4) the station card, ten one-action steps, an allergen box, safety reminders, four visible "done looks like" checks, passed by the paired station's read; (5) the cooked meal, served inside the period with a signed station check; (6) the four-question reflection with the cost comparison from the real receipt.

Steps by day: Lesson 6.12 (day 158) launch, constraints, five costed options, the choice and the justification, seven roles. Lesson 6.13 Day 1 (day 159) the scale factor, every line multiplied, unit conversions, the yield check. Lesson 6.13 Day 2 (day 160) costing from the circular, the four running totals on the wall, the swaps, the peer check, the signature. Lesson 6.14 (day 161) sticky notes into eight blocks, the critical path, the station card, the paired read, the class master grid and the collision fixes in red. Lesson 6.15 (day 162) the read-aloud consolidation, one class list by store section, unit prices and swaps, the twelve-item approval checklist, the signature. Lesson 6.16 Day 1 (day 163) Stations 1 and 2 cook while 3 and 4 prep, serve, and run the clean check. Lesson 6.16 Day 2 (day 164) reversed. Lesson 6.17 (day 165) 30 covers set, the final course served others-first, four clean zones, the waste count, the cost comparison, the reflection, the folder in.

Group or individual: the station team, about seven students at four stations or nine to ten at three. Seven project roles for the planning days (Station Manager, Menu Lead, Scale Lead, Cost Lead, Flow Lead, Shopping Lead, Reflection Lead) and the room's standard lab roles on cook day (Station Manager, Head Cook, Second Cook, Prep Cook, Helper Cook, Plate Builder, Time Keeper, plus Toppings Chief at eight). Nobody holds the same job in both.

Choice built in: the dish, from five costed options plus a sixth the team may propose and price itself; every store-brand and size swap; the whole flowchart, including which shortcut to take if the plan slips; the station card format, written or drawn as the picture version; the reflection format, written, typed, dictated, recorded, or drawn as a marked-up flowchart; and the language of the host's opening sentence at the table.

ELL and IEP supports: the picture flowchart card (eight drawn boxes with symbols) counts as a real station card; sentence starters for the choice, the swap log, the flowchart,

and the reflection in Spanish, all to be verified by a native speaker; a calculator for every student on every day; a pre-filled scaling table with the factor already entered and a four-page circular excerpt with only this project's items; the first two rows of the costing sheet and the consolidated list worked as examples; a fixed physical position for every lab role, written on the duty strip and kept in the student's pocket; Plate Builder, Time Keeper, and the prep station as burner-free roles available to anyone on both cook days at no rubric cost; a written lab alternative for a student absent on a cook day. Full list in Project 06, Supports.

How it is presented: the meal itself is the presentation. Thirty covers set properly, the allergen box read to the class, the class served before the cooks serve themselves, and a real receipt projected on the wall while students compute what they actually spent.

Full no-kitchen version: the meal becomes a no-cook assembled meal (a wrap bar, a snack board, overnight oats, a salsa and bean side, or the dessert cups), five options priced for 28 portions at station scale, and the rest of the project is unchanged: the same eight days, the same six deliverables, the same budget, the same rubric, the same flowchart with assembly phases instead of cook phases, the same station card, the same pre-flight and duty list, the same 30 covers, the same four clean zones, the same reflection. What is lost is the second performance of FSP 1. e) with heat. What is gained is a harder cold-chain and hold-time problem. Say that out loud so the no-kitchen version does not read as the consolation prize. Full menu and prices in Project 06.

9. Assessment plan

- **Formative (daily):** the do now every day, a named check for understanding in every lesson (clipboard walk with a yes or no per station, a spoken key check with a right answer, a whole-class wall chart, thumbs, a quick sort), an exit card or a one-line share most days, the Time Keeper's actual-times sheet on both cook days, and the peer checks in Lessons 6.13 and 6.14. All daily work is scored complete, partial, or missing.
- **Formal at the end of each topic:**
 - **Topic 6.1:** Assessment 06.1 - Food Systems Quiz.md, 12 items, 21 points, in the last 15 minutes of Lesson 6.5 Day 2. Covers the supply chain, local and seasonal, waste, food deserts, and sustainable methods.
 - **Topic 6.2:** Assessment 06.2 - Cooking Methods Check.md, 12 items, 21 points, in the last 15 minutes of Lesson 6.11. Covers dry heat and moist heat, the leavening experiment, and the place setting. The three labs in this topic (6.7, 6.8, 6.9) are each scored separately on Rubric 01 - Lab Rubric.md (16 points) in the labs and projects category; this check covers the knowledge, not the cooking.
 - **Topic 6.3:** Rubric 06 - Feed the Class.md, five criteria, 20 points, scored from the project folder collected at the end of Lesson 6.17 and returned at the start of Lesson 6.18 Day 2. Cook day is also scored individually on Rubric 01 - Lab Rubric.md.
 - **Topic 6.4:** no separate check; assessed inside the unit test.
- **Unit test:** Assessment 06.3 - Unit 6 Test.md, 20 items, 40 points, in the last 20 minutes of Lesson 6.18 Day 2, after a 15-minute team review game. Matching, multiple

choice, three short answers, two math items (scaling and unit price), one place setting item, one scenario, and one written response. A grade 6 version omits items 19 and 20 and is scored out of 26. One retake after a review, best score counts.

- **Grade weights within the unit** follow the course grading plan: labs and projects 40 percent (three lab rubrics at 16 points each, the cook day lab rubric, and Rubric 06 at 20 points), topic assessments 25 percent (the Food Systems Quiz, the Cooking Methods Check, and the unit test), daily work 20 percent (do nows, handouts, exit cards, the project's working papers), participation and safety 15 percent (role performance in four labs and two cook days; a safety violation is a zero for that day's participation and a conversation, plus a redo of the clean phase).
- **Marking period note:** the whole unit sits inside marking period 4 (days 136 to 180). Every check in this unit lands in MP4.
- **Missed days:** a student who misses a lab or a cook day does the written lab alternative under the grading plan (a step-by-step analysis of their own station card) and is scored on Rubric 01 Objectives 1 to 3 out of 12, scaled to 16, and on Rubric 06 criteria 1, 2, 3, and 5 with their team. A student who misses a quiz or the test takes it during the next do now or in a support period.

10. Materials, budget, and room

Room assumptions: a FACS lab with three or four kitchen stations of about seven students each, a stove with at least four burners, an oven, a microwave, a sink per station or two shared sinks, a refrigerator, wall space for four station posters and the class master flowchart, and tables students can set as a dining room. A projector. No knives beyond butter knives and kitchen shears.

Consumables, per section, in the order the unit spends them:

Line	Lesson	What it buys	Cost	One time or every section
Kitchen scale (grams and ounces)	6.3	The waste audit's measurement. The tally version needs no scale	\$12 to \$15	One time , and it is used again in 6.7 and 6.17
Potting mix, seeds, cups	6.5	The windowsill planting, alive through Lesson 6.11	about \$11	Mix and cups one time ; seeds about \$3 every year
White vinegar	6.7 Day 1	The acid and base fizz test; a pantry item afterward	about \$2	One time per year
Quick bread lab,	6.7 Day 2	Flour, milk,	about \$10 to	Every section

four stations		eggs, oil, leavening, muffin liners; about \$2.50 to \$3.50 per station	\$14	
Cheesy Pasta Marinara lab, four stations	6.8	Pasta, marinara, mozzarella, parmesan; about \$9.76 per station at full prices, about \$6.32 with store-brand swaps	about \$39 (\$25 with swaps)	Every section
Black Bean Quesadillas lab, four stations	6.9	Tortillas, black beans, cheese blend, salsa, spray; about \$12.94 per station at full prices, about \$9.50 with swaps. The optional dessert pushes a station over the \$20 cap, so run it as one shared class platter instead	about \$52 (\$38 with swaps) plus about \$8 if the dessert platter runs	Every section
Global foods tasting (optional)	6.10 Day 2	Three packaged flatbreads, labels kept in the room	about \$8 to \$10	Every section it runs
Place settings, paper and plastic	6.11	Only if the room does not own place settings; these are reused in Lessons 6.16 and 6.17	about \$8	One time
Feed the Class	6.12 to 6.17	The approved	about \$80 (\$75	Every section

project food		consolidated shopping list: four station dishes for 28 portions plus 2 spares. \$20 per station, \$80 for the class	to \$78 in practice after swaps)	
Paper goods for the project	6.16, 6.17	30 plates, forks, spoons, cups, napkins per cook day, if the room is short	about \$8	Every section
Printing	6.1 to 6.18	About 16 pages per student across the unit, plus four poster-size flowchart blanks and five menu option cards per class	school printing	Every section

Per-section total at full prices: about \$211. With the store-brand swaps on the two labs: **about \$180.** Take out the one-time lines (the scale, the potting mix and cups, the vinegar, the place settings, about \$36) and a second or third section of the same course costs about **\$175 at full prices, about \$145 with swaps.**

Across four sections in one year that is roughly **\$600 to \$700**, and the single largest line is the project at \$80 per section. That is the number to get approved before a class is promised a meal. If the budget will only fund half, three stations instead of four with the two cheapest options brings a section's project to about \$55.

Equipment (buy what is missing): four large pots, four colanders (teacher use only), four sets of measuring cups and spoons, eight mixing bowls, eight sheet pans, four timers, four pairs of oven mitts per station, two calibrated thermometers, one kitchen scale, sanitizing spray and cloths for every station, four bus tubs, a compost bucket if the school composts, painter's tape, and a class set of about 30 basic calculators (about \$30 one time; used every day of Topic 6.3 and again in Unit 7).

Handouts (in 05 Materials/Unit 06/, as the lessons name them):

- Topic 6.1: Handout 06.01 - Farm to Plate Path Cards.md; Handout 06.02 - New York Seasonal Chart and Two Meals Task.md; Handout 06.03 - Waste Audit Data Sheet and Scale Up.md; Handout 06.04 - Neighborhood Map and Price Comparison.md; Handout 06.05 - Compost Sort and Packaging Redesign.md

- Topic 6.2: Handout 06.06 - Method Match and Demonstration Record.md; Handout 06.07 - Quick Bread Lab Sheet.md; Handout 06.08 - Station Card Cheesy Pasta Marinara.md; Handout 06.08 - Role Cards Lab 3.md; Handout 06.09 - Station Card Black Bean Quesadillas.md; Handout 06.10 - Dish Research Card and Comparison.md; Handout 06.11 - Place Setting Diagram and Sharing Agreement.md
- Topic 6.3: Handout 06.12 - Menu Option Cards and Constraint Card.md; Handout 06.13 - Scaling Worksheet and Costing Sheet.md; Handout 06.14 - Flowchart Builder and Station Card Template.md; Handout 06.15 - Consolidated Shopping List and Approval Checklist.md; Handout 06.16 - Cook Day Duty List and Pre-Flight Check.md; Handout 06.17 - Serve and Clean Zones and Project Reflection.md
- Topic 6.4: Handout 06.18 - Food Systems Careers Cards.md

Slides: one outline per lesson, Slides 06.01 through Slides 06.18, in the same folder. The site build renders every outline as an HTML deck.

No-kitchen plan for the whole unit: every lesson in Topic 6.1 and Lessons 6.6, 6.10, 6.11, 6.12, 6.13, 6.14, 6.15, and 6.18 already run without a kitchen. Lesson 6.7's quick bread has a no-oven version; Lesson 6.8 becomes a Cold Noodle Bowl (about \$11 per station) and Lesson 6.9 becomes Rolled Bean and Cheese Wraps (about \$13 per station), both on the same roles, phases, and station check. The project's full no-cook version is in Project 06. A whole-unit no-kitchen run costs slightly more in paper goods and slightly less in food, and it teaches the cold chain better than the kitchen version does.

11. Health and other-department coordination

Science. Two places, and both are real opportunities rather than courtesies. **The leavening lab (6.7)** is a controlled experiment with one variable and four constants; the science teacher's words for variable, control, hypothesis, and conclusion should be the words on the lab sheet, so ask which wording the grade 7 and grade 8 science classes use and match it. **The growing in 6.5** is a plant life cycle with light, water, and soil as the inputs, and the cups sit on the windowsill for six weeks, which is longer than most science units can hold a living thing. FACS gives science a four-batch data set with a measurable outcome and a windowsill; science gives FACS the vocabulary and, if willing, five minutes on Day 2 of Lesson 6.7 to look at the four batches.

Math. Lesson 6.13 is ratio, unit conversion, and unit rate doing real work, and Lesson 6.15 is unit price. Use the same setup and the same words the math class uses: the scale factor as a ratio, the price per ounce as a unit rate, and the variance in Lesson 6.17 as percent change (end minus start over start). FACS gives math four real costing sheets, a set of unit-rate problems with a store circular attached, and the class's own cost-per-person number; math gives FACS the method students already know and, if willing, ten minutes on Lesson 6.13 Day 2 to check the four running totals.

The cafeteria manager. Three things, and this is the most useful relationship in the unit. **The waste audit (6.3):** the audit is more honest with the manager's cooperation, and the

manager may already have a number. **The allergy list:** the building already runs an allergen protocol and the manager knows it; the class allergy list for Lessons 6.7 through 6.17 should match what the cafeteria works from, and the nurse signs off on it. **Purchasing (6.15):** five minutes on how the school actually buys food, what a case costs against a retail package, and what the school's real per-meal figure is. FACS gives the manager a class that can read a unit price and a purchase list and a waste number from their own room; the manager gives FACS the building's numbers and its protocol.

Health. This unit does not teach nutrition. Unit 1 taught the six nutrients, MyPlate, and labels, and the health course owns dietary guidance, body image, and eating disorders. Unit 6's overlap with health is three narrow things: food safety and the danger zone (already taught in Unit 1 and practiced here), food access and food insecurity (Lesson 6.4, where the health course may also cover community health resources), and the cultural and social side of eating (Lessons 6.10, 6.11, 6.17). Anything a student says in Lesson 6.4 about food insecurity at home is a quiet referral to the counselor, not a class discussion, and the lesson says so.

Social studies. If the social studies course covers trade routes, migration, or the movement of crops between continents, Lesson 6.10 is the same content from the plate side. Share the dish research card. FACS stops at what people eat and why; the causes and the chronology belong to social studies.

ELA. The station card in Lesson 6.14 is procedural writing with a test attached, and the reflection in Lesson 6.17 is a claim with evidence. Share both templates so students see one structure for a procedure and one for an argument.

12. Connections (in-building and community)

Inside the building:

- **The cafeteria manager**, for Lessons 6.3, 6.15, and 6.17. **FACS gives:** the waste number from the class's own audit and again from the project's Zone 4 count, a class that can read a unit price and a purchase list, and thirty students who now know what the kitchen staff actually do. **FACS gets:** the building's real per-meal cost, the allergen protocol the school already runs, five minutes on case pricing versus retail pricing, and cooperation on the audit.
- **The science teacher**, for Lessons 6.5 and 6.7. **FACS gives:** a four-batch controlled experiment with a measurable rise, a windowsill of living plants that lasts six weeks, and a lab sheet using science's own vocabulary. **FACS gets:** the department's wording for variable, control, and hypothesis, and, if willing, five minutes in the room on the day the batches come out of the oven.
- **The math teacher**, for Lessons 6.13 and 6.15 (see Section 11).
- **The custodian and the building engineer**, for Lessons 6.16 and 6.17. A two-minute conversation the week before about a four-burner, full-oven day and a room that gets cleaned to a signed checklist, and in exchange a working exhaust fan, an extra bus tub, and a warning if the hot water is out.

Community:

- **A supermarket manager** (the store the circular comes from), for Lessons 6.12, 6.13, and 6.15. **FACS gives:** a class of customers who can read a unit price, a thank-you note from every student, and a real purchase. **FACS gets:** a stack of this week's circulars, a five-minute answer on why a store brand costs less and who writes the circular, sometimes a donation toward the \$80, and occasionally permission to bring two students along on the shopping trip.
- **A Cornell Cooperative Extension educator from the county office**, for Lessons 6.5, 6.4, and 6.18. **FACS gives:** a class that has audited its own waste, planted something, and studied food access, plus a ready audience for a program Extension already runs in schools. **FACS gets:** a professional who can talk about local growing, composting, food safety, and nutrition, materials that are free and already written for this age, and a real person behind the farmer, agricultural engineer, and dietitian cards. This is the easiest yes on the list.
- **A local farm or farmers market vendor** on the North Fork or at the nearest market, for Lessons 6.2 and 6.18. **FACS gives:** twenty-eight families who now know what "in season" means and where the market is, and a thank-you from the class. **FACS gets:** ten minutes on what actually grows here and when, what a grower is paid compared to the shelf price, and one story about a season that went wrong.
- **A food pantry or food rescue organization**, for Lessons 6.4 and 6.17. **FACS gives:** unopened, in-date, shelf-stable leftovers where the pantry's rules allow it, a class that has counted its own waste, and students who know what SNAP and WIC are. **FACS gets:** one visit or one email answering what they wish more people understood about feeding a group, and real numbers about need in this county.

13. Vocabulary for translation

See Vocabulary 06.md: 49 Tier 3 terms across the four topics with a one-line student definition written for a grade 6 to 7 reader and a Spanish column, to be checked by a native speaker, a bilingual colleague, or the ENL teacher. The file also carries sentence starters in English and Spanish for three things: taking a role in a lab, writing the dish research card, and writing the project reflection. Each lesson names its own Tier 3 words and its handout carries a word bank pulled from the table. The site's Materials page prints the table as picture-ready cards.